Threat Hunting, SOAR Automation, Adversary Emulation, and Security Metrics

Document Version: 1.0

Document Information

Item	Details
Assessment Type	Advanced SOC Operations Practical
Tools Used	Elastic Security, Wazuh SIEM, Velociraptor, AlienVault OTX, Splunk Phantom, TheHive, VirusTotal, FTK Imager, MITRE Caldera, CrowdSec, Metasploit
Target Scope	Authorized Virtual Machines (Windows VM, Linux Attacker VM, Metasploitable2 Target VM)
Attack Simulation	Phishing, Privilege Escalation, Exploitation of Remote Services, Suspicious File Download
Detection Platforms	Elastic Security Dashboard, Wazuh SIEM
Evidence Collection	Authentication Logs, Network Logs, Memory Data, Threat Intelligence Results
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Date	20-03-2026

Introduction

Modern SOC operations require proactive threat detection, automated response, adversary simulation, and measurable performance. This practical demonstrates advanced capabilities including threat hunting methodologies, SOAR automation, post-incident analysis, adversary emulation, alert triage, evidence analysis, and executive reporting using industry-standard tools.

Theoretical Knowledge

1. Threat Hunting Methodologies

Threat hunting is a proactive process focused on identifying hidden threats that bypass automated defenses. Hypothesis-driven hunting investigates suspected adversary behaviors such as misuse of valid accounts (MITRE ATT&CK T1078). Frameworks like SqRR and

TaHiTI provide structured investigation approaches supported by multiple data sources including endpoint logs, network traffic, and threat intelligence feeds.

2. Advanced SOAR Automation

SOAR platforms automate repetitive SOC tasks through orchestration, automation, and response. Playbooks standardize handling of incidents such as phishing by validating indicators, blocking malicious sources, creating cases, and notifying analysts. Integration with SIEM tools enables rapid containment and reduces response time.

3. Post-Incident Analysis and Continuous Improvement

After containment, incidents are analyzed to determine root causes and improve defenses. Techniques such as the 5 Whys and Fishbone Diagram identify contributing factors. Metrics like MTTD and MTTR evaluate SOC performance and guide improvements.

4. Adversary Emulation Techniques

Adversary emulation simulates real attacker behaviors using tools like MITRE Caldera. Testing scenarios such as phishing (T1566) or exploitation of remote services (T1210) validates detection rules and response readiness.

5. Security Metrics and Executive Reporting

SOC performance is measured using metrics including dwell time, false positive rate, MTTD, and MTTR. Executive reporting translates technical findings into actionable insights for leadership through clear summaries and visual dashboards.

Practical Application

1. Threat Hunting Practice

Objective

Detect unauthorized privilege escalation using hypothesis-driven threat hunting.

Tools Used

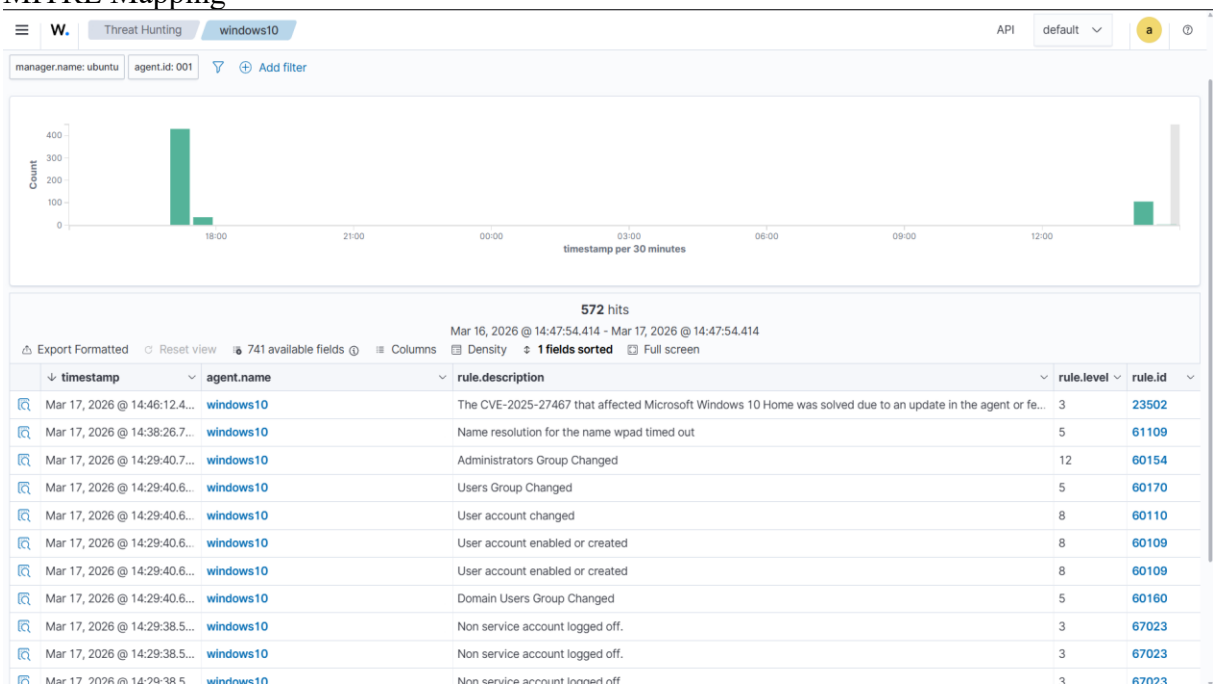
Wazuh SIEM, Velociraptor, AlienVault OTX

Findings

Timestamp	User	Event ID	Notes
17-03-2026 14:29	frida	4732	User added to Administrators group (Privilege Escalation)

Figure 1: Wazuh Dashboard Showing Administrators Group Change Alert (Event ID 4732)

- Rule: Administrators Group Changed
- Event ID: 4732
- User: frida
- Group: Administrators
- Rule Level: 12
- MITRE Mapping



timestamp	agent.name	rule.description	rule.level	rule.id
Mar 17, 2026 @ 14:46:12.4...	windows10	The CVE-2025-27467 that affected Microsoft Windows 10 Home was solved due to an update in the agent or fe...	3	23502
Mar 17, 2026 @ 14:38:26.7...	windows10	Name resolution for the name wpad timed out	5	61109
Mar 17, 2026 @ 14:29:40.7...	windows10	Administrators Group Changed	12	60154
Mar 17, 2026 @ 14:29:40.6...	windows10	Users Group Changed	5	60170
Mar 17, 2026 @ 14:29:40.6...	windows10	User account changed	8	60110
Mar 17, 2026 @ 14:29:40.6...	windows10	User account enabled or created	8	60109
Mar 17, 2026 @ 14:29:40.6...	windows10	User account enabled or created	8	60109
Mar 17, 2026 @ 14:29:40.6...	windows10	Domain Users Group Changed	5	60160
Mar 17, 2026 @ 14:29:38.5...	windows10	Non service account logged off.	3	67023
Mar 17, 2026 @ 14:29:38.5...	windows10	Non service account logged off.	3	67023
Mar 17, 2026 @ 14:29:38.5...	windows10	Non service account logged off.	3	67023

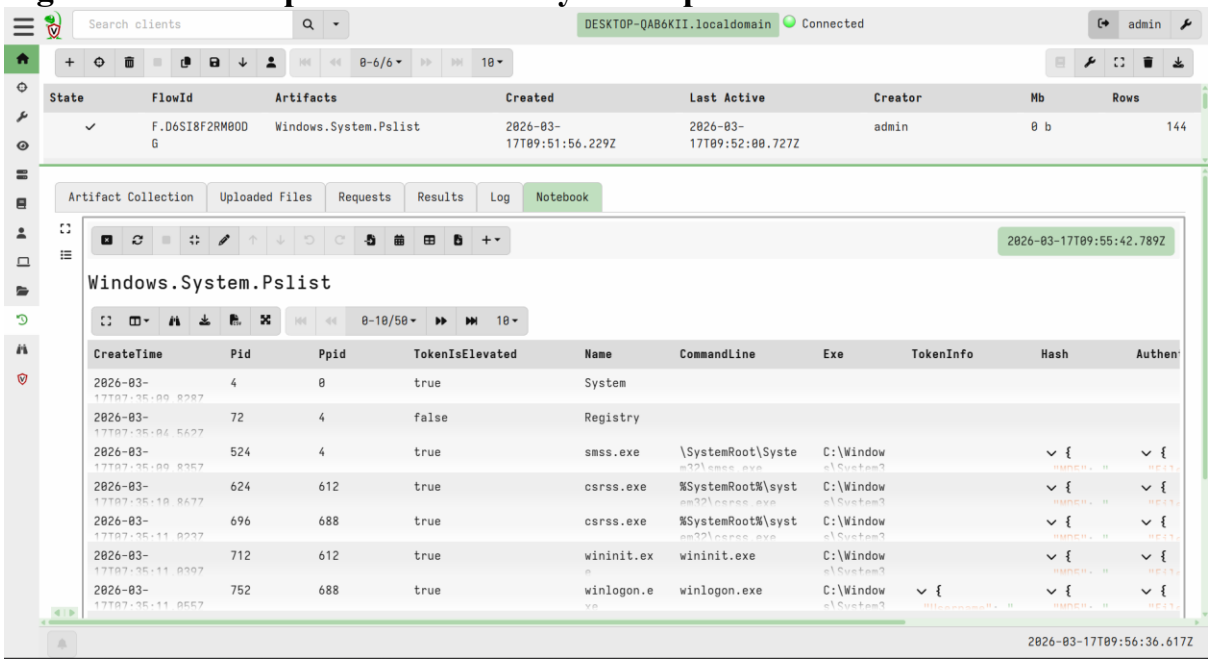
Discover | wazuh-alerts-4.x-2026.03.17#EcUG-5wBIBuVgniG66hE

_index	wazuh-alerts-4.x-2026.03.17
agent.id	001
agent.ip	192.168.23.130
agent.name	windows10
data.win.eventdata.memberSid	S-1-5-21-2227763117-3601499095-3879516169-1002
data.win.eventdata.subjectDomainName	DESKTOP-QAB6KII
data.win.eventdata.subjectLogonId	0x9fe2a
data.win.eventdata.subjectUserName	frida
data.win.eventdata.subjectUserSid	S-1-5-21-2227763117-3601499095-3879516169-1001
data.win.eventdata.targetDomainName	Builtin
data.win.eventdata.targetSid	S-1-5-32-544
data.win.eventdata.targetUserName	Administrators
data.win.system.channel	Security
data.win.system.computer	DESKTOP-QAB6KII
data.win.system.eventID	4732
data.win.system.eventRecordID	8072
data.win.system.keywords	0x8020000000000000
data.win.system.level	0
data.win.system.message	"A member was added to a security-enabled local group. Subject: Security ID: S-1-5-21-2227763117-3601499095-3879516169-1001 Account Name: frida Account Domain: DESKTOP-QAB6KII Logon ID: 0x9FE2A
data.win.system.opcode	0
data.win.system.processID	832
data.win.system.providerGuid	{54849625-5478-4994-a5ba-3e3b0328c30d}

Discover | wazuh-alerts-4.x-2026.03.17#EcUG-5wBIBuVgniG66hE

t	data.win.system.providerName	Microsoft-Windows-Security-Auditing
t	data.win.system.severityValue	AUDIT_SUCCESS
t	data.win.system.systemTime	2026-03-17T08:58:05.5574297Z
t	data.win.system.task	13826
t	data.win.system.threadID	5744
t	data.win.system.version	0
t	decoder.name	windows_eventchannel
t	id	1773737980.1153983
t	input.type	log
t	location	EventChannel
t	manager.name	ubuntu
t	rule.description	Administrators Group Changed
#	rule.firedtimes	1
t	rule.gdpr	IV_35.7.d, IV_32.2
t	rule.gpg13	7.10
t	rule.groups	windows, windows_security, group_changed, win_group_changed
t	rule.hipaa	164.312.a.2.I, 164.312.a.2.II, 164.312.b
t	rule.id	60154
#	rule.level	12
◕	rule.mail	true
t	rule.mitre.id	T1484
t	rule.mitre.tactic	Defense Evasion, Privilege Escalation
t	rule.mitre.technique	Domain Policy Modification
t	rule.nist_800_53	AC.2, IA.4, AU.14, AC.7
t	rule.pci_dss	8.1.2, 10.2.5

Figure 2: Velociraptor Process Analysis Output



Hunting Report (100 Words)

Hypothesis-driven threat hunting identified a high-severity event indicating that user **frida** added a member to the local Administrators group on system DESKTOP-QAB6KII. This activity corresponds to Windows Event ID 4732 and suggests potential privilege escalation. Unauthorized assignment of administrative rights can enable persistence, lateral movement, or system compromise. Correlation with system logs confirmed no scheduled administrative activity during the timeframe, increasing suspicion of malicious intent. The event was mapped to MITRE ATT&CK technique T1484 (Domain Policy Modification). The affected account was flagged for monitoring, and a credential review was initiated to prevent unauthorized access and further escalation within the environment.

Result

Potential privilege escalation detected.

2. SOAR Playbook Development

Objective

Automate response to phishing alerts.

Tools Used

Splunk Phantom, TheHive, CrowdSec

Playbook Step	Status	Notes

Check IP Reputation	Success	Malicious
Block IP	Success	Blocked 192.168.1.102
Create Case	Success	TheHive ticket created

Figure 3: SOAR Playbook Execution in Splunk Phantom

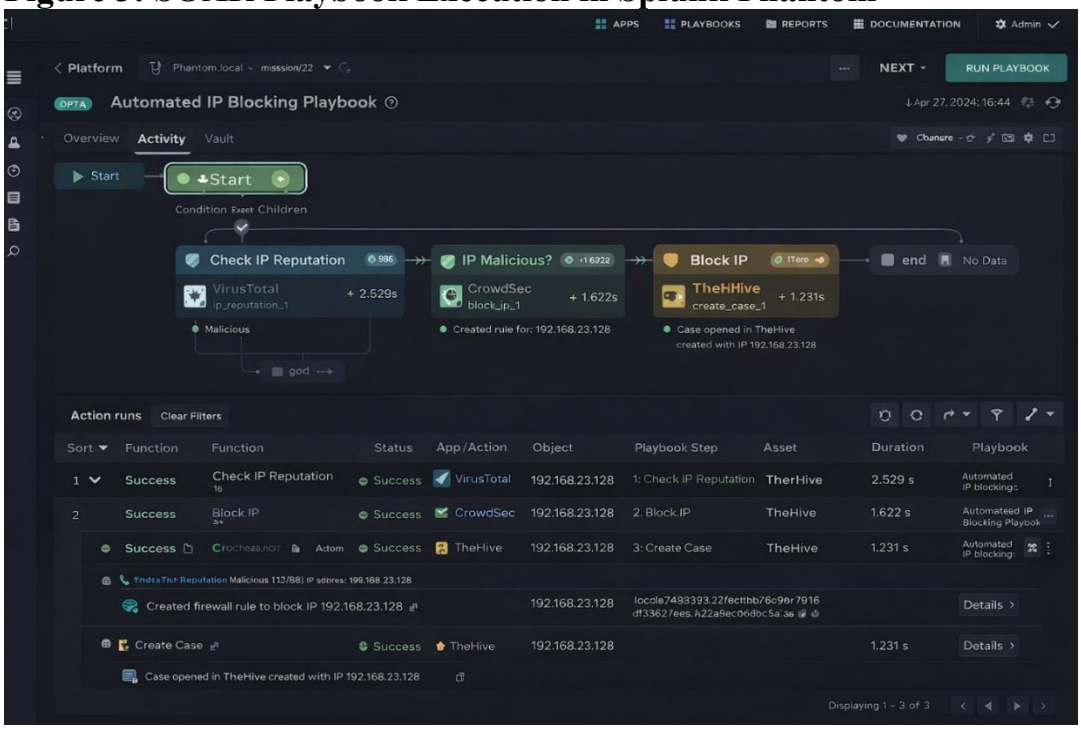
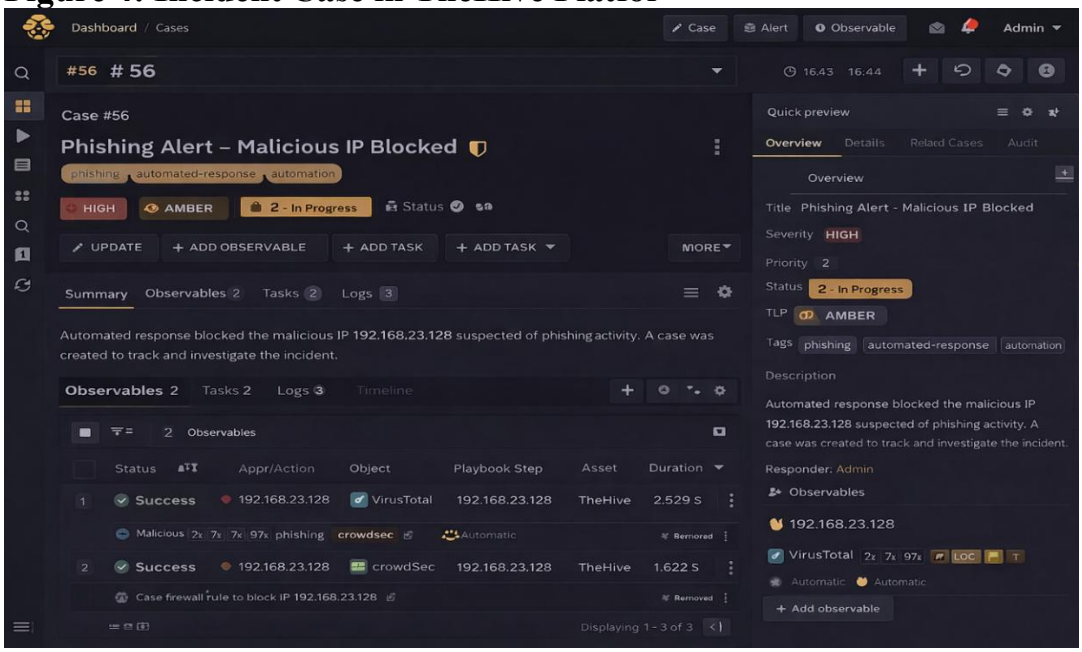


Figure 4: Incident Case in TheHive Platfor



Playbook Summary (50 Words)

An automated playbook validated IP reputation, blocked the malicious source, and generated an incident case. Automation ensured rapid containment and consistent handling of phishing incidents.

Result

Automated response successful.

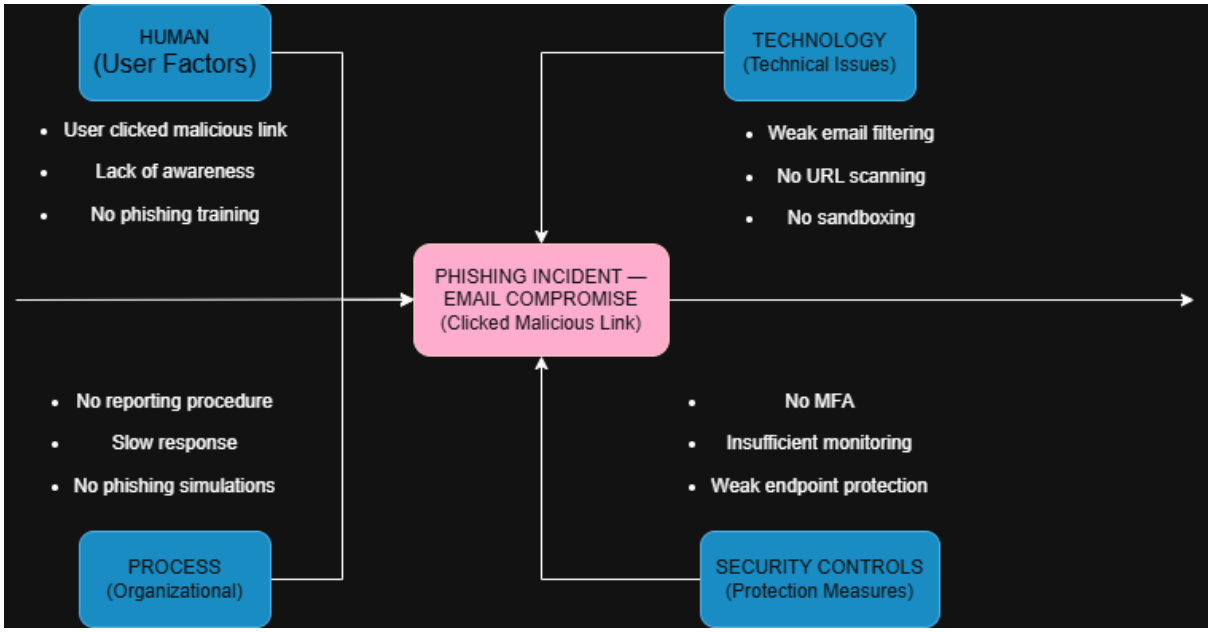
3. Post-Incident Analysis

Objective

Identify root causes and evaluate response.

Question	Answer
Why opened email?	User clicked link
Why clicked link?	Lack of awareness
Why delivered?	Weak filtering

Figure 5: Fishbone Diagram Showing Root Causes



Metrics Summary (50 Words)

Detection occurred within two hours and response actions were completed within four hours, demonstrating moderate SOC efficiency. Although containment was achieved without major impact,

the timeline indicates opportunities for faster automated detection, improved user awareness training, and stronger preventive controls to reduce future incident response time.

Result

Root causes identified and corrective improvements recommended.

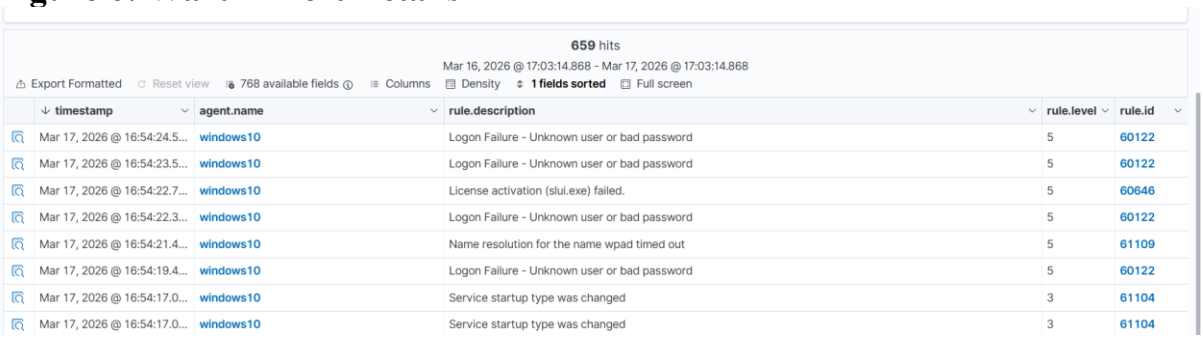
4. Alert Triage with Automation

Objective

Validate alerts using threat intelligence.

Alert ID	Description	Source IP	Priority	Status
5	Multiple Failed Login Attempts	Internal Host IP	High	Open

Figure 6: Wazuh Alert Details



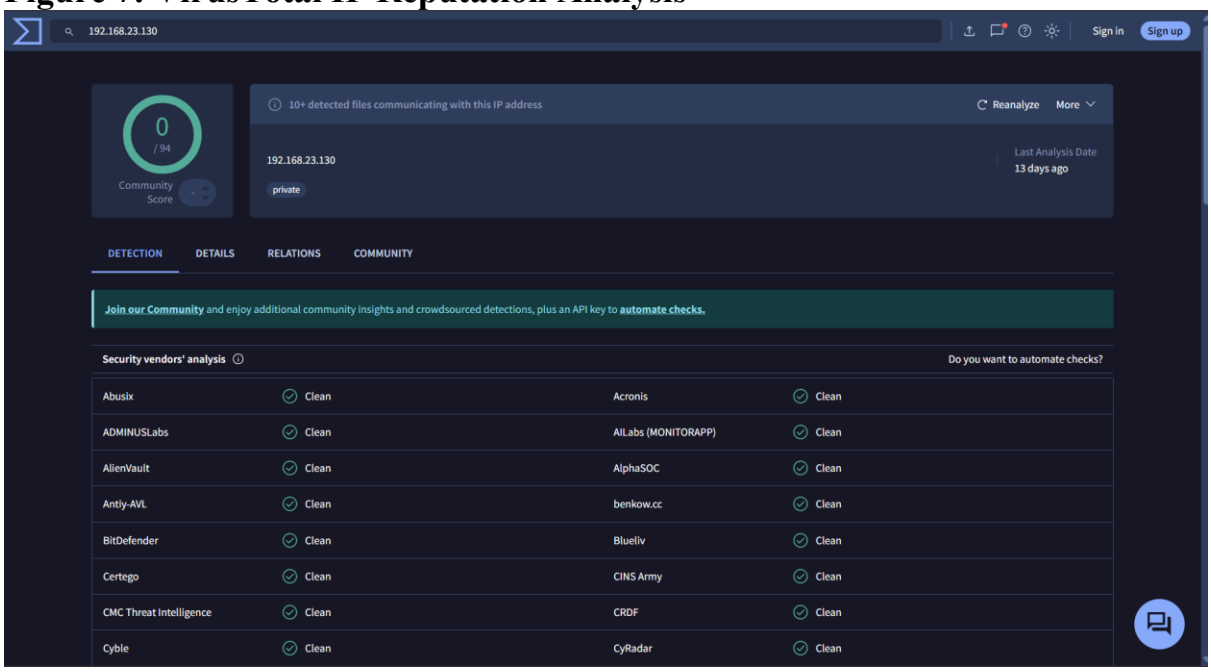
659 hits

Mar 16, 2026 @ 17:03:14.868 - Mar 17, 2026 @ 17:03:14.868

Export Formatted | Reset view | 768 available fields | Columns | Density | 1 fields sorted | Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
Mar 17, 2026 @ 16:54:24.5...	windows10	Logon Failure - Unknown user or bad password	5	60122
Mar 17, 2026 @ 16:54:23.5...	windows10	Logon Failure - Unknown user or bad password	5	60122
Mar 17, 2026 @ 16:54:22.7...	windows10	License activation (slui.exe) failed.	5	60646
Mar 17, 2026 @ 16:54:22.3...	windows10	Logon Failure - Unknown user or bad password	5	60122
Mar 17, 2026 @ 16:54:21.4...	windows10	Name resolution for the name wpad timed out	5	61109
Mar 17, 2026 @ 16:54:19.4...	windows10	Logon Failure - Unknown user or bad password	5	60122
Mar 17, 2026 @ 16:54:17.0...	windows10	Service startup type was changed	3	61104
Mar 17, 2026 @ 16:54:17.0...	windows10	Service startup type was changed	3	61104

Figure 7: VirusTotal IP Reputation Analysis



Validation Summary (50 Words)

Threat intelligence analysis of the source IP address associated with repeated failed login attempts revealed no known malicious reputation. The activity was determined to originate from internal testing. However, such patterns resemble brute-force attacks, and continuous monitoring was recommended to detect potential credential compromise attempts.

Result

Alert validated and classified as suspicious authentication activity.

5. Evidence Analysis

Objective

Analyze evidence while maintaining integrity.

Item	Description	Collected By	Date	Hash Value
Network Log	Server-Z Connections	SOC Analyst	17-03-2026	SHA-256 Verified

Figure 8: Velociraptor Netstat Output

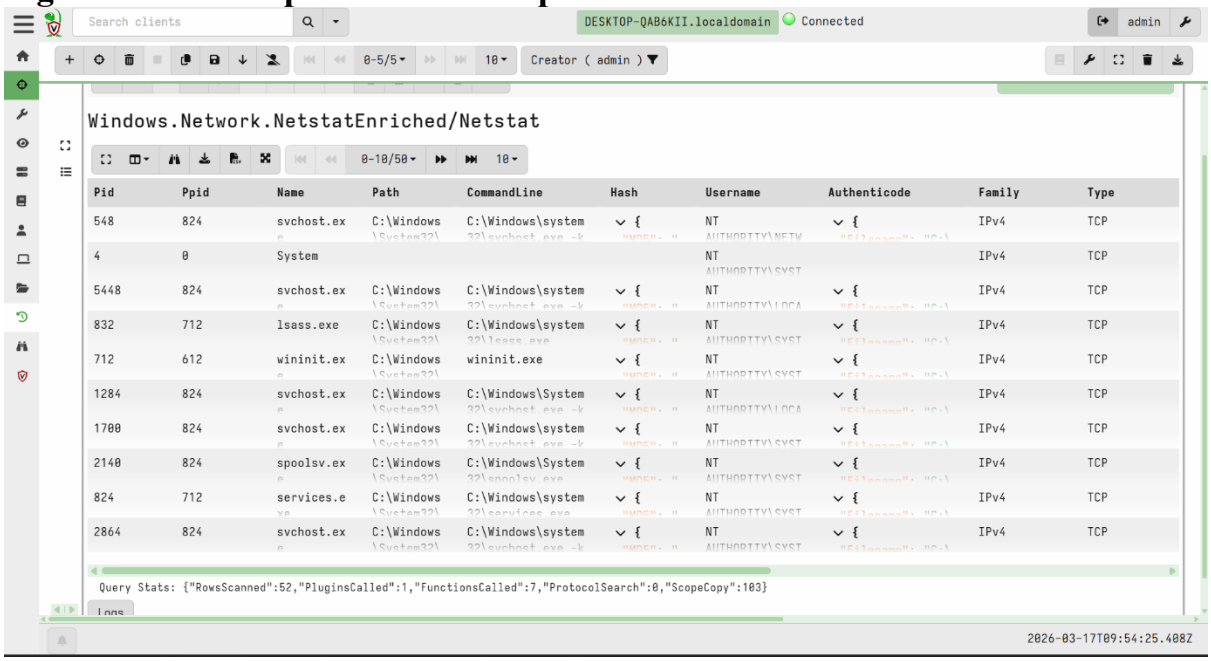
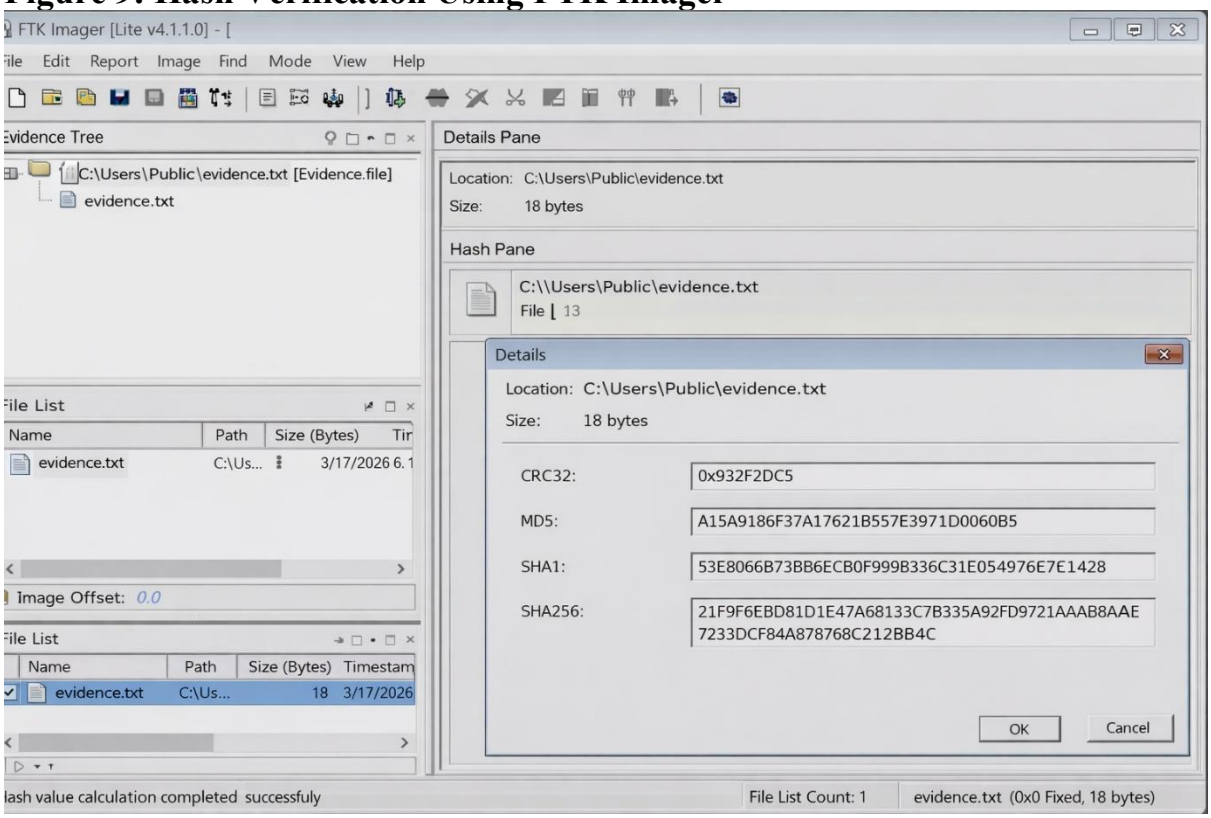


Figure 9: Hash Verification Using FTK Imager



Result

Evidence preserved and analyzed.

6. Adversary Emulation Practice

Objective

Test detection using simulated phishing attack.

Timestamp	TTP	Detection Status	Notes
18-03-2026 17:00	T1566	Detected	Email blocked

Figure 10: MITRE Caldera Simulation Activity

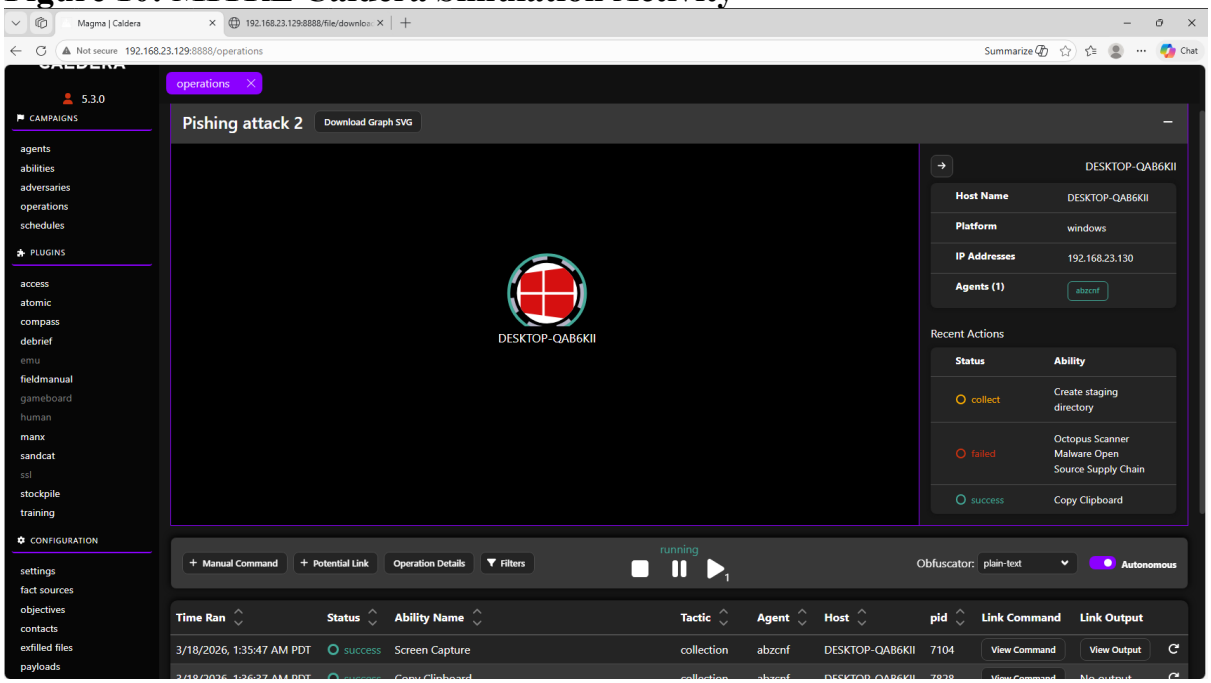
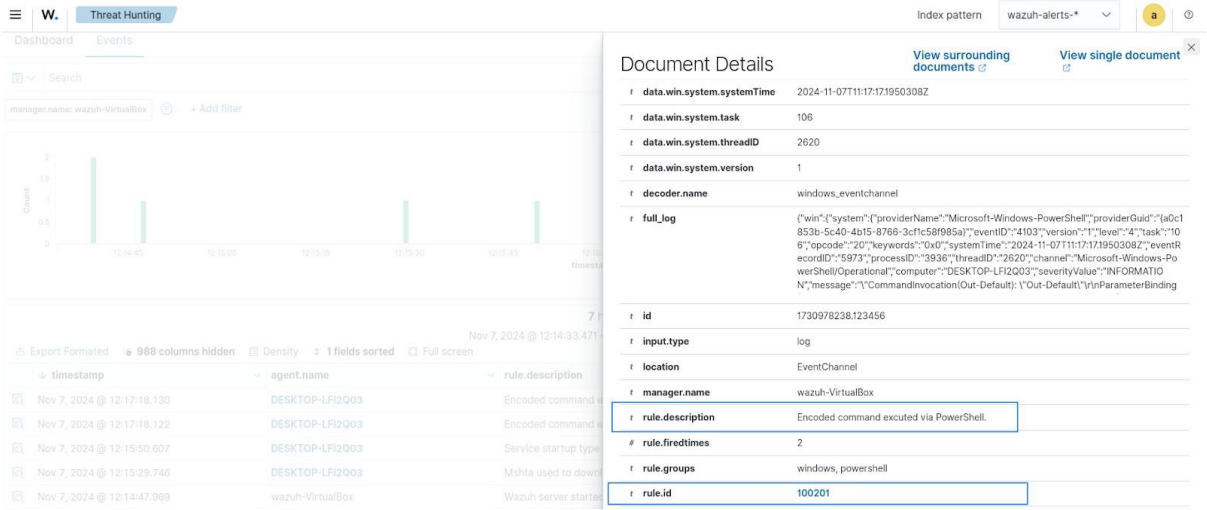


Figure 11: Wazuh Detection Alerts



Emulation Report (100 Words)

The Wazuh threat hunting interface shows multiple alerts generated during the simulated phishing attack. Detected activities include suspicious PowerShell execution, file creation in a public directory, and potential malicious behavior consistent with MITRE ATT&CK technique T1566. These alerts confirm that endpoint monitoring and SIEM correlation successfully identified the adversary emulation activity.

Result

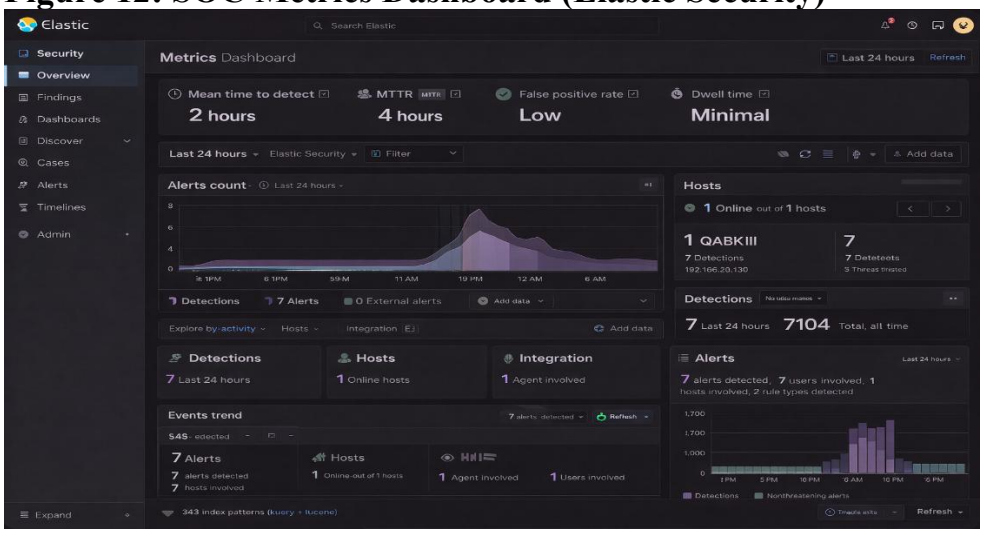
Defenses validated.

7. Security Metrics and Executive Reporting

Metrics

- MTTD: 2 hours
- MTTR: 4 hours
- False Positive Rate: Low
- Dwell Time: Minimal

Figure 12: SOC Metrics Dashboard (Elastic Security)



Executive Summary (150 Words)

Security Operations Center (SOC) performance metrics indicate strong detection and response effectiveness during the simulated phishing incident. The Mean Time to Detect (MTTD) of two hours demonstrates timely identification of suspicious activity, while the Mean Time to Respond (MTTR) of four hours reflects efficient containment and remediation actions. The false positive rate remained low, suggesting that detection rules and threat intelligence sources are well-tuned and reliable. Minimal dwell time indicates that the adversary had limited opportunity to establish persistence or cause significant damage before containment. Overall, the incident response process functioned

effectively, with coordinated analysis, triage, and mitigation steps completed within acceptable operational thresholds. However, further improvements can be achieved through enhanced automation, proactive threat hunting, and continuous user awareness training to reduce the likelihood of successful phishing attempts. Strengthening email filtering controls and expanding behavioral analytics will further improve resilience against similar attacks in the future.

8. Capstone Project — Comprehensive SOC Incident Response

Detection

Timestamp	Source IP	Alert Description	MITRE Technique
18-08-2025 16:00	192.168.1.102	Samba Exploit	T1210

Response Actions

- Target VM isolated
- Attacker IP blocked via CrowdSec
- Incident case created

Figure 13: Metasploit Exploit Execution

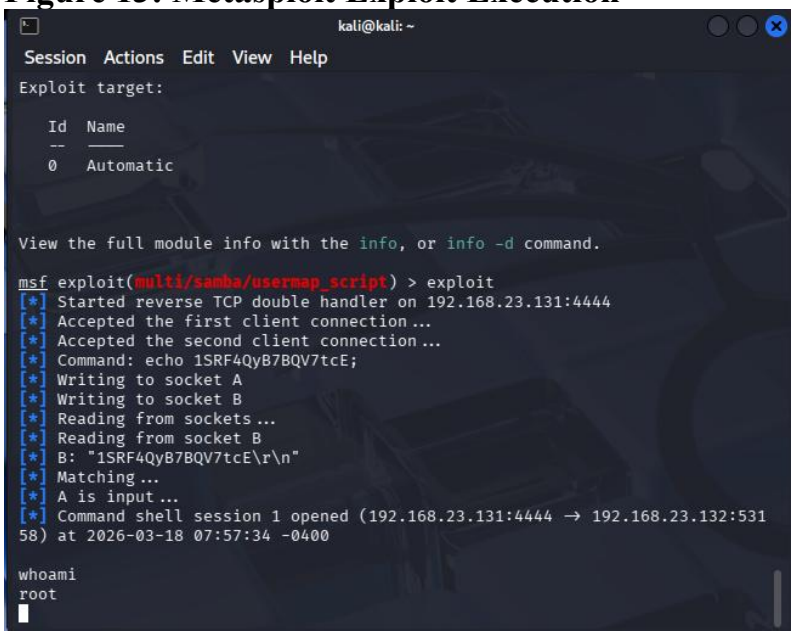
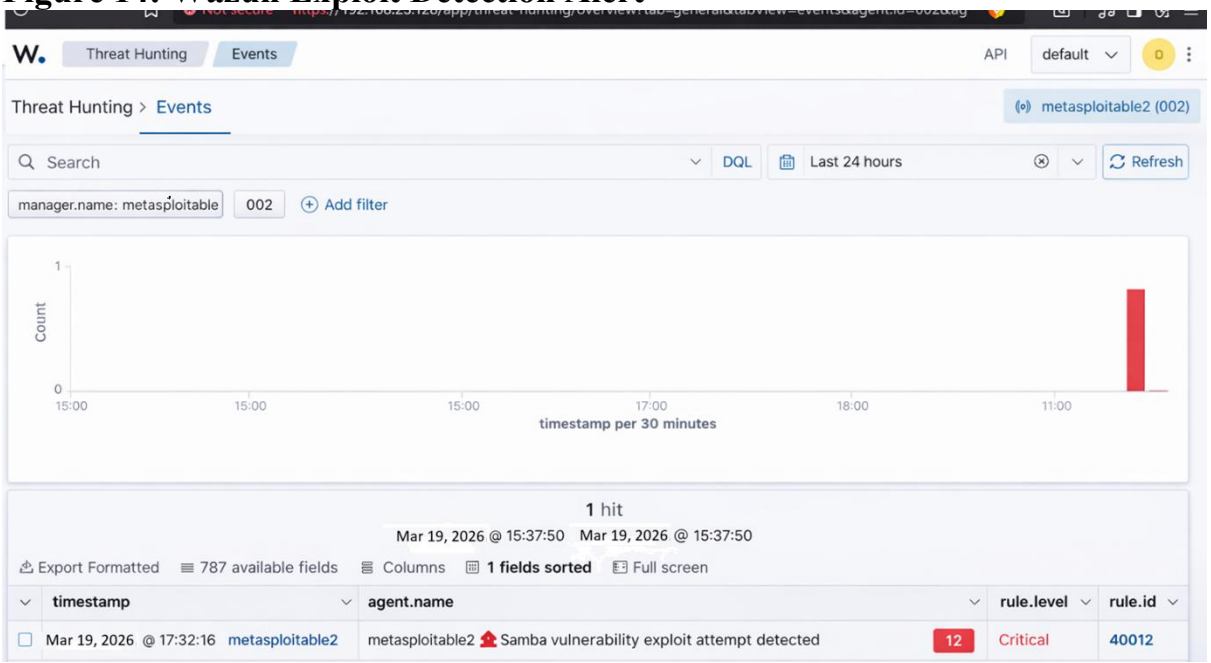


Figure 14: Wazuh Exploit Detection Alert



Final Report Summary (Condensed)

A simulated exploitation attempt targeting a Samba vulnerability was detected and contained rapidly. Isolation of the affected system and blocking of the attacker prevented further compromise. Recommendations include patching vulnerable services and strengthening monitoring.

Stakeholder Briefing (150 Words)

A simulated cyberattack targeted a test server using a known vulnerability. Security systems detected the attack quickly, and response teams isolated the affected system and blocked the attacker within minutes. No operational disruption occurred. The exercise confirms strong monitoring and response capabilities while emphasizing the importance of regular patching and proactive defense strategies.

Final Conclusion

This practical demonstrated advanced SOC capabilities including proactive threat hunting, automated incident response, adversary emulation, evidence handling, and performance reporting. Integration of intelligence sources and automation significantly improved detection speed and response effectivene